

Support Vector Machines (SVM) – Geometric Intuition

1 What is SVM?

- SVM is a **supervised learning algorithm** mainly used for **classification** (sometimes regression).
 - Goal: Find the **best decision boundary (hyperplane)** that separates data into classes.
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2 Geometric Intuition

Imagine you have two classes of points on a 2D plane (say red circles and blue squares). Many possible lines (boundaries) can separate them, but SVM looks for the **best one**.

Best = the line (or hyperplane in higher dimensions) that maximizes the margin.

3 What is Margin?

- **Margin** = the distance between the decision boundary and the **closest data points** from each class.
 - SVM chooses the boundary with the **maximum margin**.
 - Why? → A larger margin = more confidence in predictions, better generalization to unseen data.
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4 Support Vectors

- The data points that lie **closest to the hyperplane** (i.e., those that define the margin).
 - They are the **most critical points** → moving or removing them changes the boundary.
 - Other points farther away don't affect the decision boundary.
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5 SVM in Higher Dimensions

- In 2D → boundary is a **line**.
 - In 3D → boundary is a **plane**.
 - In n-dimensions → boundary is a **hyperplane**.
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6 Non-linear Data

- If data is not linearly separable → SVM uses the **kernel trick**.
 - Kernel transforms data into a higher dimension where a linear separation becomes possible.
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- ❖ Example: Data in a circle → kernel maps it into higher space where a plane separates it.

7 Why SVM Works Well

1. Maximizes margin → robust to noise.
2. Uses only **support vectors**, not all points → efficient.
3. Flexible → with kernels, it handles complex boundaries.

➤ Quick Recap

- **SVM = Best separating hyperplane with maximum margin.**
- **Margin = distance from boundary to closest points.**
- **Support Vectors = points that define the margin.**
- **Kernel trick = handles non-linear data.**

👉 Geometrically, think of SVM as stretching a rubber band between two classes and finding the **widest possible gap** where the band sits.
